

Naga Jayadeep Narla

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PROFESSIONAL SUMMARY

Data Scientist & Machine Learning Engineer with an M.Sc. in Commercial Vehicle Technology (RPTU Germany) and a strong foundation in Mechanical Engineering. Experienced in building production-ready MLOps pipelines, deep learning models for computer vision, and time-series forecasting in industrial environments across research roles at DFKI and dual internships at Bosch. Proficient in Python, SQL, PyTorch, Docker, MLflow, with strong mathematical expertise in vector calculus, linear algebra, and statistical modeling.

CORE COMPETENCIES & TECHNICAL SKILLS

- Languages & Querying:** Python (NumPy, Pandas, Scikit-Learn, PyTorch), SQL (PostgreSQL, MSSQL), C++ (Intermediate)
- Machine Learning & Deep Learning:** Time-Series Forecasting, Convolutional Variational Autoencoders (CVAE), Random Forest Regressors, Computer Vision, Instance Segmentation, Object Detection, Image Processing
- MLOps & Infrastructure:** Docker, Docker-Compose, MLflow, Prefect, Flask, Fast API, Evidently AI, Grafana, Git/GitHub, CI/CD
- Tools, Databases & OS:** Linux (Ubuntu), VS Code, Jupyter Notebook, PowerBI, Grafana, Matlab/Simulink
- Domain Expertise:** Industrial Automation (Industry 4.0), Robot Grasp Synthesis, Predictive Maintenance, Sensor Signal Decoding (CAN bus)

WORK EXPERIENCE & RESEARCH

DFKI & RPTU Kaiserslautern

Kaiserslautern, Germany

Master Thesis: Optimal Grasping Strategy for work tool handover by cobots(CV&Robotics) **Aug 2024 – Feb 2025**

- Researched, designed, and trained end-to-end deep learning architectures (object detection and instance segmentation) on multi-GPU clusters for real-time cobot grasp synthesis.
- Built and annotated custom high-resolution image datasets, executing systematic data filtering and augmentation strategies to prevent model overfitting.
- Fine-tuned deep learning hyperparameters to maximize bounding box precision, mask segmentation and grasping rectangle generation for complex industrial work tools. An overall 68.75% of successful grasps across various tools have been achieved.
- Developed real-time camera inference scripts, converting 2D and 3D pixel coordinates into robot-space 3D matrices using rigid body transformation calculations.
- Evaluated real-time model latency and memory footprints to ensure smooth hardware execution on cobot embedded platforms.

Bosch

Stuttgart, Germany

Data Science & Machine Learning Intern (Prognostic Health Monitoring)

May 2023 – Sep 2023

- Decoded and parsed complex multivariate time-series sensor signals from Controller Area Network (CAN) bus protocols using DBC files for predictive maintenance tasks.
- Assisted in developing deep learning architectures using Convolutional Variational Autoencoders (CVAE) to detect early sensor anomalies and predict health deterioration in electronic components.
- Engineered modular Python preprocessing pipelines featuring strict error handling, automated logging, and data imputation to preserve critical time dependencies in continuous sensor data.
- Created real-time data visualizers and interactive plotting utilities to monitor feature distributions and signal drift during pipeline runtime.
- Documented end-to-end data processing workflows using detailed architectural flowcharts and maintained code quality through collaborating with other developers using Git/GitHub.

Bosch

Homburg, Germany

Data Science Intern (Industry 4.0 & Manufacturing Data)

Nov 2022 – Jan 2023

- Extracted, cleaned, and structured high-frequency time-series data from smart factory assembly lines to predict key quality parameters (bore diameter tolerances).
- Engineered domain-specific features from raw sensor signals and conducted automated feature selection using correlation matrices and tree-based importance scores.

- Trained Random Forest Regressors using Grid Search CV and 5-fold cross-validation, achieving significant improvements in prediction accuracy over baseline statistical models.
- Designed dashboards to visualize model residual errors and feature importance distributions for manufacturing plant engineers.

KEY PROJECTS & PORTFOLIO

End-to-End Production MLOps Pipeline (Iron Ore Quality Prediction)

May 2025 – Jul 2025

DataTalksClub MLOps Zoomcamp

- Engineered a production-grade machine learning pipeline to predict iron ore quality parameters using continuous real-world industrial processing data.
- Standardized model experiment tracking and version management using **MLflow**, maintaining a centralized model registry for seamless deployment transitions.
- Orchestrated end-to-end training and inference workflows using **Prefect**, establishing automated scheduling and data validation.
- Containerized model deployment using **Docker** and **Docker-Compose**, serving real-time prediction endpoints via a lightweight **Flask** API web service.
- Integrated continuous monitoring tools using **Evidently AI** to detect data drift, logging metric telemetry into a **PostgreSQL** database and visualizing alerts on interactive **Grafana** dashboards.

Technical Blog & Applied Data Science Notes

Jun 2026 – Present

Personal Technical Portfolio

- Authored technical breakdowns covering fundamental mathematical concepts in Data Science, focusing on vector calculus (Jacobians, Hessians), matrix decomposition, and probability distributions.
- Published step-by-step implementations of exploratory data analysis (EDA), hypothesis testing, and statistical sampling distributions in Python.

EDUCATION & CERTIFICATIONS

M.Sc. in Commercial Vehicle Technology

2020 – 2025

RPTU Kaiserslautern-Landau, Germany

- *Core Focus:* 2D/3D Computer Vision, Deep Learning, Bus Systems, Real-Time Systems

Certification in Data Science and AI

Oct 2022 – May 2023

IIT Madras & Intellipaat

- Mastered RDBMS concepts and SQL execution (DDL, DML, DCL, TCL) on MSSQL Server for database management and analytical querying.
- Applied supervised and unsupervised machine learning algorithms, deep learning models, and PowerBI visualization workflows to complex datasets.

B.Tech. in Mechanical Engineering

2016 – 2019

Acharya Nagarjuna University, India

LANGUAGES

English (Professional Working Proficiency) · **German** (B1 Level) · **Telugu** (Native)